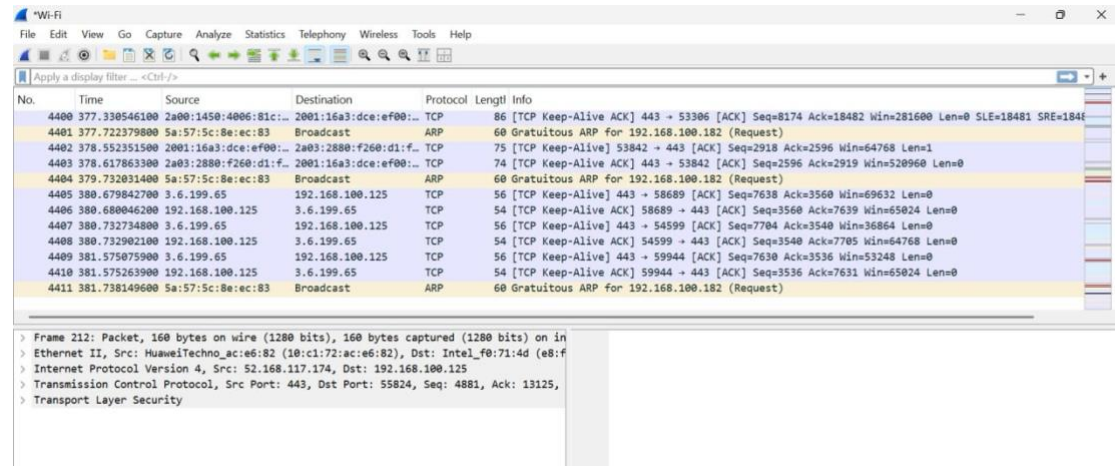


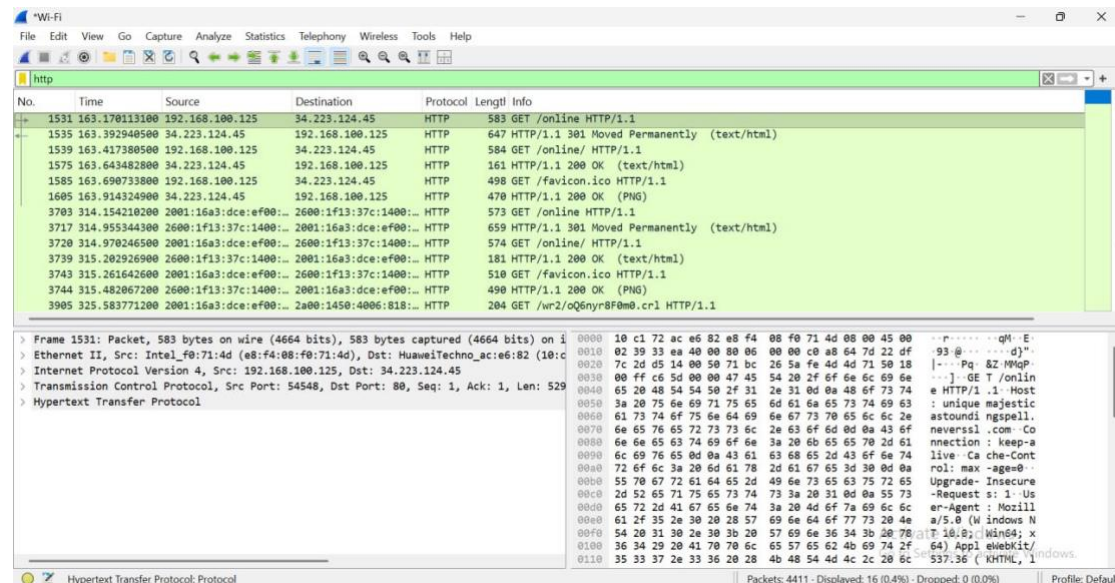
CS471 - LAB 1

Part 1: Capturing HTTP Traffic.

Task 1: Start Wireshark and capture packets.

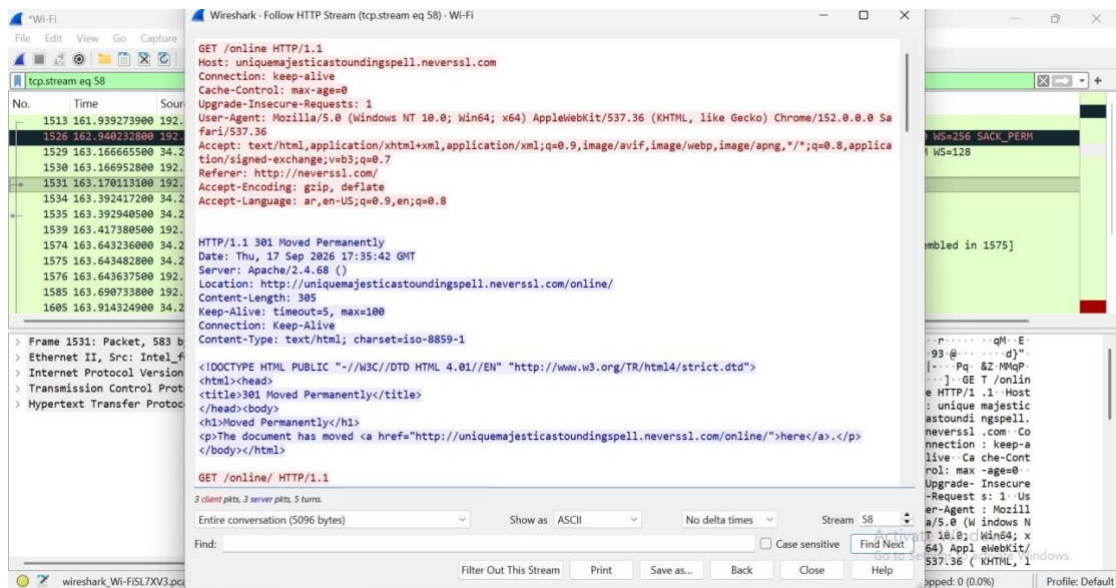


Task 2: Filter HTTP packets and analyze them.



Part 2: Analyzing TCP/IP Traffic.

Task 1: Filter TCP packets

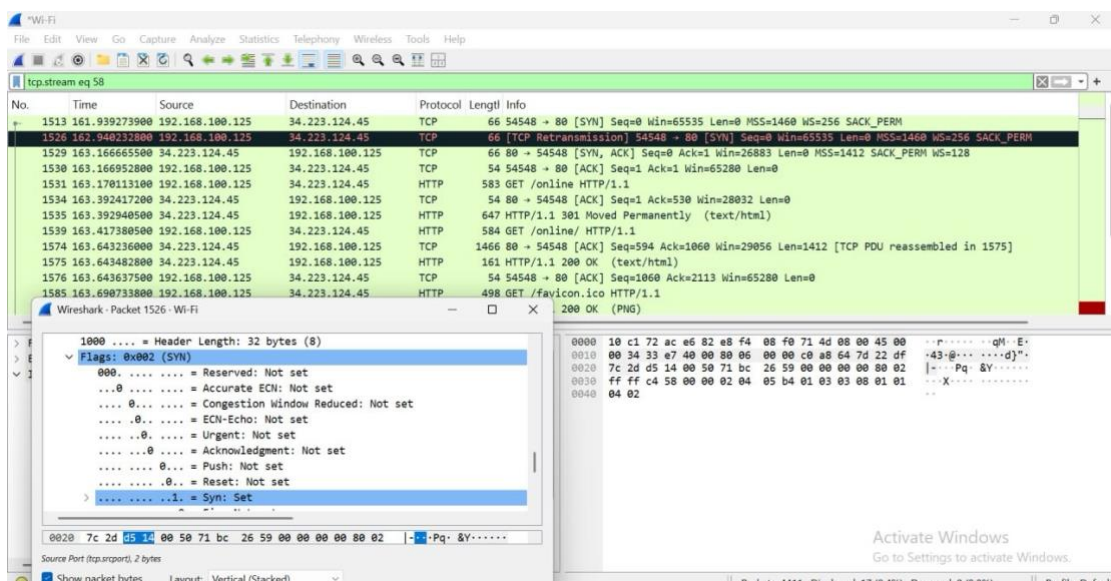


Task 2: Analyze TCP handshake and investigate Data Transfer and Termination.

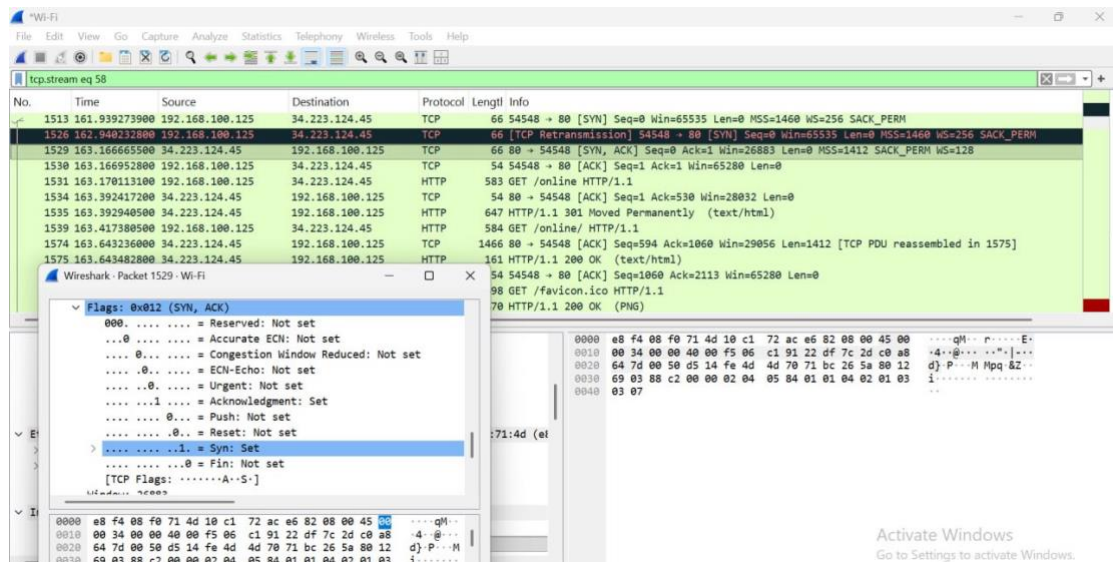
Step 1: Find and select packets related to the TCP three-way handshake:

Step 2: Note the sequence and acknowledgment numbers.

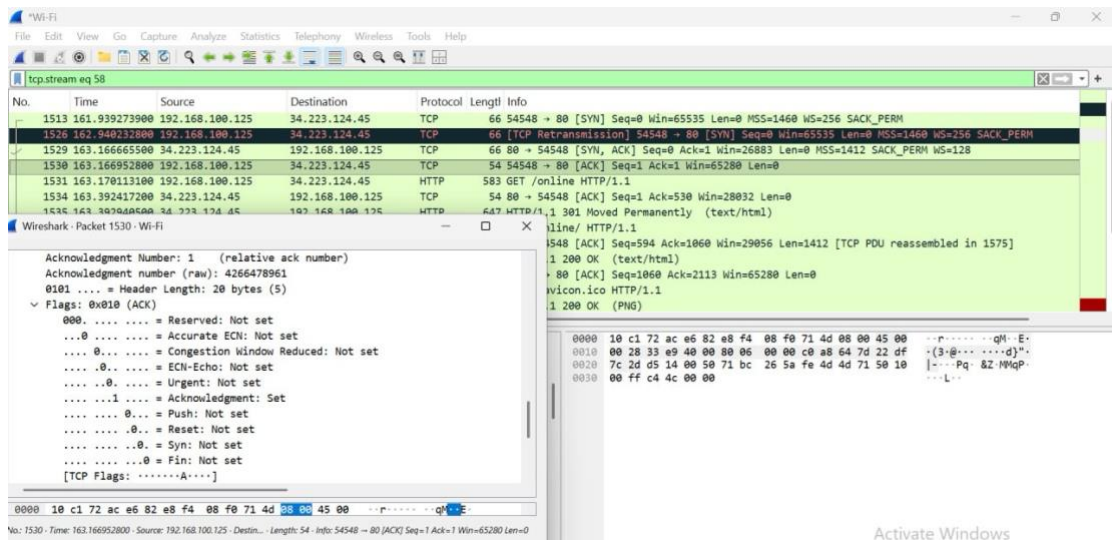
SYN: Initiates a connection.



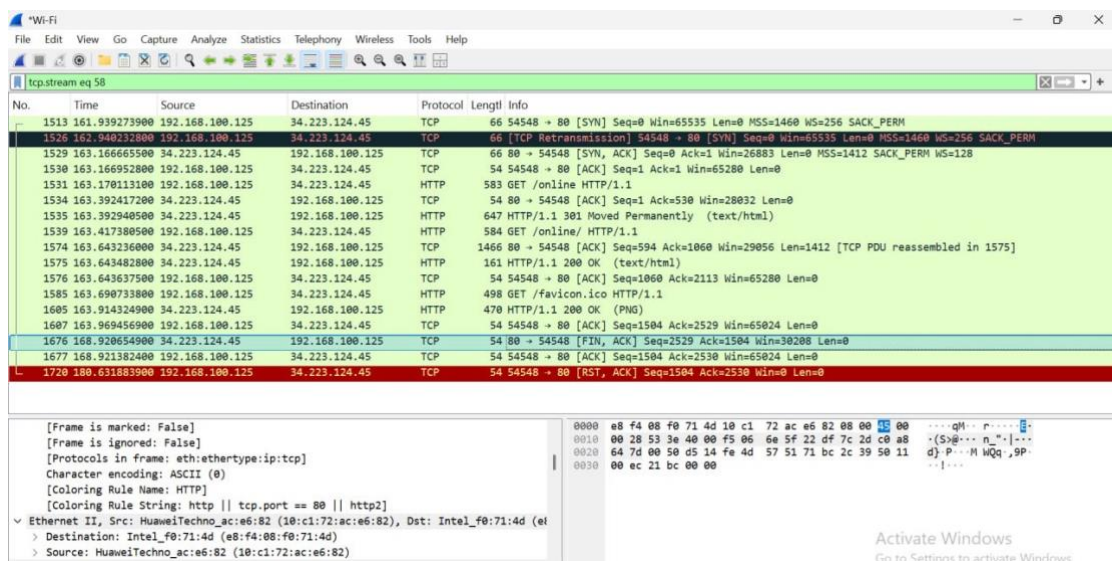
SYN-ACK: Acknowledges and responds to the SYN.



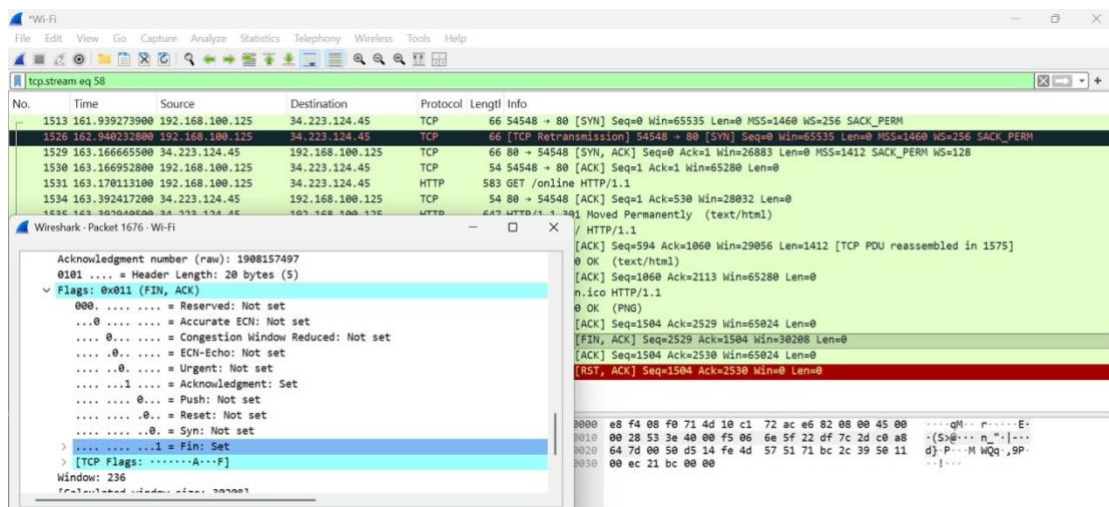
ACK: Acknowledges the SYN-ACK and establishes the connection.



Step 3: Observe the data packets exchanged between the client and server.



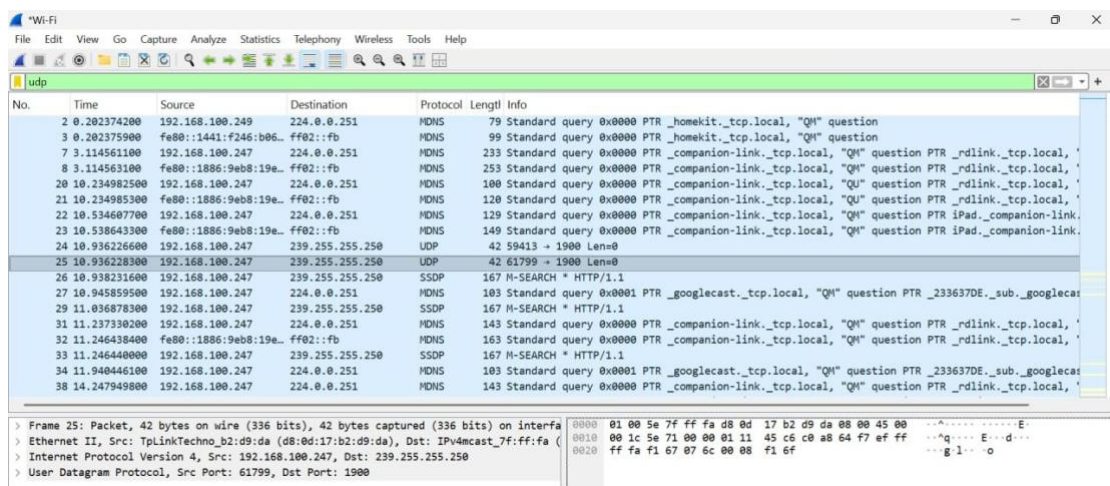
Step 4: Look at the TCP termination process (FIN, ACK packets).

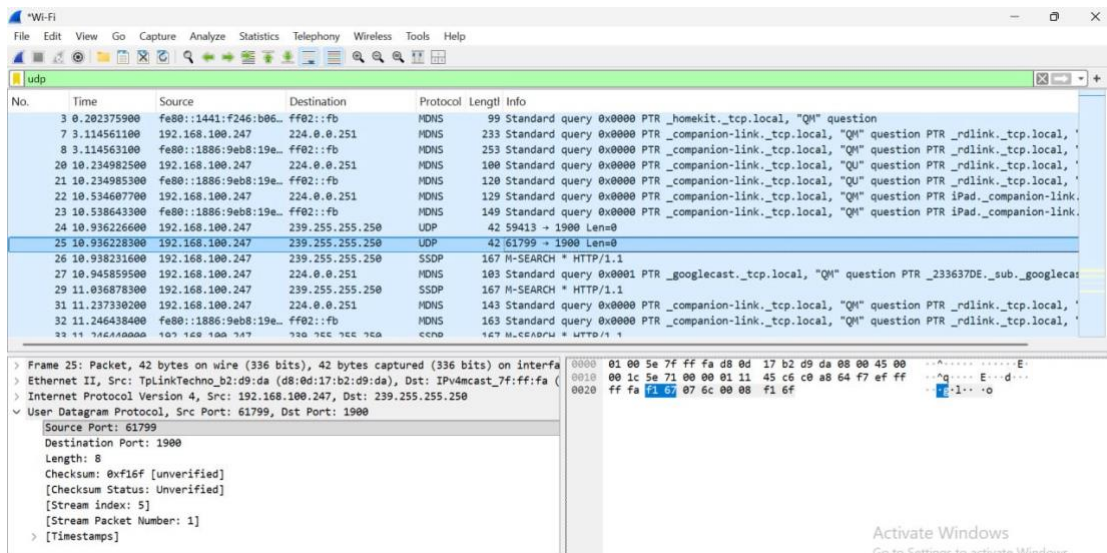


Part 3: Capturing and Analyzing UDP Traffic

Task 1: Generate UDP traffic and capture packets.

Task 2: Filter and analysis UDP Packets





The TCP header is larger because it contains many fields to manage and guarantee the connection.

The UDP header is smaller and simpler, making it suitable for speed and lightweight communications.

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	connection-oriented and reliable
Data Integrity and Ordering	TCP	Ensure correct and ordered delivery

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	web browsing, Email, File transfer	streaming, online gaming, VoIP
Performance	Slower but reliable	Faster but less reliable

